

Test yourself

Q1: Write each of the following in matrix form and solve using Gauss-Jordan reduction:

- a. $2.x_1 - 3.x_2 = 2$ and $-4.x_1 + 2.x_2 = 3$
- b. $x_1 + x_2 = 1$, $x_2 + x_3 = 1$ and $2.x_1 + 3.x_2 - x_3 = 0$

Q2: Hoy et al, p299, Q1

For each of the matrix operations below, identify those that result in a scalar?

- a) AB where A is a 2×1 and B is a 1×2
- b) $A^T B$ where A is a 2×1 and B is a 2×1
- c) $A^T B A$ where A is a 2×1 and B is a 2×2
- d) $AA^T B$ where A is a 5×1 and B is a 5×5

Q3: Let

$$A = \begin{bmatrix} 1 & -1 & 3 \\ 0 & -2 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 1 & 0 \\ -3 & 2 \end{bmatrix}, C = \begin{bmatrix} -1 & 1 & -3 \\ 0 & 2 & -2 \end{bmatrix} \text{ and } D = \begin{bmatrix} -1 & -2 & 0 \\ 1 & -2 & 1 \\ 2 & 1 & 4 \end{bmatrix}$$

Calculate each of the following if defined: $A+3B$, $A+C$, $C+2D$, AB , BA , CD , DC

Q4: Hoy et al, p299, Q2

Show that $AB \neq BA$ where:

$$A = \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix}$$

Q5: Hoy et al, p299, Q3

Let

$$A = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} \text{ and } x = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Compute $(Ax)^T$, $x^T A^T$, xx^T , and $x^T x$. Is $A^T x^T$ defined?

Q6: Hoy et al, p345, Q1

Does the following matrix have an inverse?

$$\begin{bmatrix} 12 & 13 & 14 \\ 15 & 16 & 17 \\ 18 & 19 & 20 \end{bmatrix}$$

Q7: Determine the lengths of the column vectors:

$$u = \begin{bmatrix} 1 \\ 2 \\ -2 \\ 1 \end{bmatrix}, v = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \text{ and } w = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

Are any of these vectors mutually perpendicular?

Q8: Hoy et al, p345, Q3

Use the co-factor expansion method to compute $|A|$, where:

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

Q9: Hoy et al, p345, Q5

Use Cramer's rule to compute the solutions to

$$2x_1 + x_2 = 3$$

$$-3x_1 + x_3 = -8$$

$$x_2 + 2x_3 = -2$$

Q10: Hoy et al, p388, Q1

Establish whether the following pairs of vectors are linearly independent:

a) $y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

b) $y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

c) $y = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$, $w = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $z = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$

Q11: Hoy et al, p388, Q3

- a) If A is a 9×7 matrix, what is the largest possible rank?
- b) If A is a 7×9 matrix, what is the largest possible rank?

Q12: Hoy et al, p388, Q7

Find the eigenvalues of

a) $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$

b) $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$

Solutions

Q1: Write each of the following in matrix form and solve using Gauss-Jordan reduction:

c. $2.x_1 - 3.x_2 = 2$ and $-4.x_1 + 2.x_2 = 3$

d. $x_1 + x_2 = 1$, $x_2 + x_3 = 1$ and $2.x_1 + 3.x_2 - x_3 = 0$

$$\begin{bmatrix} 2 & -3 \\ -4 & 2 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -3 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$R1=R1/2$$

$$\begin{bmatrix} 1 & -3/2 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$R2=-R2/4-R1$$

$$\begin{bmatrix} 1 & -3/2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 7/4 \end{bmatrix}$$

$$R1=R1-3/2R2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 + \frac{21}{8} \\ -\frac{7}{4} \end{bmatrix}$$

$$R1 = 1 + 21/8 = 3.625$$

$$R2 = -7/4 = -1.75$$

Q2: Hoy et al, p299, Q1

For each of the matrix operations below, identify those that result in a scalar?

e) AB where A is a 2×1 and B is a 1×2

f) $A^T B$ where A is a 2×1 and B is a 2×1

g) $A^T B A$ where A is a 2×1 and B is a 2×2

h) $AA^T B$ where A is a 5×1 and B is a 5×5

1. (a) AB is 2×2 .
- (b) $A^T B$ is a scalar.
- (c) $A^T B A$ is a scalar.
- (d) $AA^T B$ is 5×5 .

Q3: Let

$$A = \begin{bmatrix} 1 & -1 & 3 \\ 0 & -2 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 1 & 0 \\ -3 & 2 \end{bmatrix}, C = \begin{bmatrix} -1 & 1 & -3 \\ 0 & 2 & -2 \end{bmatrix} \text{ and } D = \begin{bmatrix} -1 & -2 & 0 \\ 1 & -2 & 1 \\ 2 & 1 & 4 \end{bmatrix}$$

Calculate each of the following if defined: $A+3B$, $A+C$, $C+2D$, AB , BA , CD , DC

$A+3B$ cannot be done

$$A+C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$C+2D$ = cannot be done

$$AB = \begin{bmatrix} -9 & 8 \\ -8 & 4 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & -5 & 7 \\ 1 & -1 & 3 \\ -3 & -1 & -5 \end{bmatrix}$$

$$CD = \begin{bmatrix} 4 & -3 & -11 \\ -2 & -6 & 7 \end{bmatrix}$$

DC not possible

Q4: Hoy et al, p299, Q2

Show that $AB \neq BA$ where:

$$A = \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 14 & 3 \\ -2 & -6 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix} \cdot \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 10 & 2 \\ 29 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 14 & 3 \\ -2 & -6 \end{bmatrix} \neq \begin{bmatrix} 10 & 2 \\ 29 & -2 \end{bmatrix}$$

Q5: Hoy et al, p299, Q3

Let

$$A = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} \text{ and } x = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Compute $(Ax)^T$, $x^T A^T$, xx^T , and $x^T x$. Is $A^T x^T$ defined?

$$3. \quad (Ax)^T = [-4 \ 2], \ x^T A^T = [-4 \ 2],$$

$$xx^T = \begin{bmatrix} 25 & 15 \\ 15 & 9 \end{bmatrix}$$

$$\text{and } x^T x = 34. \ A^T x^T \text{ is not defined.}$$

Q6: Hoy et al, p345, Q1

Does the following matrix have an inverse?

$$\begin{bmatrix} 12 & 13 & 14 \\ 15 & 16 & 17 \\ 18 & 19 & 20 \end{bmatrix}$$

1. *A has a zero determinant and therefore no inverse.*

Q7: Determine the lengths of the column vectors:

$$u = \begin{bmatrix} 1 \\ 2 \\ -2 \\ 1 \end{bmatrix}, v = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \text{ and } w = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

Are any of these vectors mutually perpendicular?

$$\|u\| = \sqrt[2]{10}$$

$$\|v\| = \sqrt[2]{2}$$

$$\|w\| = \sqrt[2]{8}$$

$v \cdot w = 0$ and hence they are perpendicular.

Q8: Hoy et al, p345, Q3

Use the co-factor expansion method to compute $|A|$, where:

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

$$3. \quad |A| = -2.$$

Q9: Hoy et al, p345, Q5

Use Cramer's rule to compute the solutions to

$$2x_1 + x_2 = 3$$

$$-3x_1 + x_3 = -8$$

$$x_2 + 2x_3 = -2$$

$$5. \quad x_1 = 11/4, x_2 = -10/4, x_3 = 1/4.$$

Q10: Hoy et al, p388, Q1

Establish whether the following pairs of vectors are linearly independent:

d) $y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

e) $y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

f) $y = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$, $w = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $z = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$

1. (a) Linearly independent, (b) linearly dependent, (c) linearly independent.

Q11: Hoy et al, p388, Q3

c) If A is a 9x7 matrix, what is the largest possible rank?

d) If A is a 7x9 matrix, what is the largest possible rank?

3. (a) Largest possible rank is 7.

(b) Largest possible rank is 7.

Q12: Hoy et al, p388, Q7

Find the eigenvalues of

c) $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$

d) $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$

7. (a) $\lambda_1 = 3, \lambda_2 = -7.$

(b) $\lambda_1 = 9, \lambda_2 = -5.$